that). We now know that Einstein's equations allow many kinds of wormholes (wormholes with many different shapes and behaviors), but Flamm's is the only one that is precisely spherical and contains no gravitating matter. When we take an equatorial slice through Flamm's wormhole, so it and our universe (our brane) have just two dimensions rather than three, and when we then view our universe and the wormhole from the bulk, they look like the left part of Figure 14.2.

With one of our universe's dimensions lost from the picture, you must think of yourself as a two-dimensional creature confined to move on the bent sheet or on the wormhole's two-dimensional wall. There are two routes for travel from location *A* in our universe to location *B*: the short route (dashed blue curve) down the wormhole's wall, or the long route (dashed red curve) along the bent sheet, our universe.

Of course, our universe is really three dimensional. The concentric circles in the left part of Figure 14.2 are really the nested green spheres shown to the right. As you enter the wormhole along the blue path from location *A*, you pass through spheres that get smaller and smaller. Then the spheres, though nested inside each other, cease changing circumference. And then, as you exit the wormhole toward location *B*, the spheres get larger and larger.

For nineteen years, physicists paid little attention to Flamm's outrageous solution of Einstein's equations, his wormhole. Then in 1935 Einstein himself and fellow physicist Nathan Rosen, unaware of Flamm's work, rediscovered Flamm's solution, explored its properties, and speculated about its significance in the real world. Other physicists, also unaware of Flamm's work, began to call his wormhole the "Einstein-Rosen bridge."